Correlating movement in space to the parameters of sound

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Abstract

This paper addresses some of the issues associated with integrating sound and body movement in space and the mapping of movement to the various parameters of sound using computers and sensors as the mediating devices.

Historically composers have made the decision as to how particular movements or trajectories through space will affect sound. This has sometimes meant that the dancers' palette is arbitrarily limited by the composer's choices or visa versa – that a choreographer's imperatives impose arbitrary limitations on the sound/music.

The question this paper addresses is whether there are any general rules or assumptions that can be made about correlating aspects of sound to aspects of movement that are at once flexible and practical and maintain integrity in both disciplines.

The paper analyses a number of examples of interactive, sensing environments and poses some possible directions from the questions above. The directions presented are supported by descriptions of objects moving in virtual space and the sound that results using a variety of mapping procedures.

Introduction

With the increase in sophistication and reliability of motion sensing equipment, more attention can now be given to compositional and choreographic concerns over the technical concerns of the instruments (Winkler, 1995). Affordability of some of the technology is still some way off. I am thinking particularly of motion capture here. But there are cheaper technical means that offer a high degree of useful motion data. Of particular interest are systems that use video to capture motion in an environment. They are relatively inexpensive and integrate with well-known programs such as MAX/MSP. Eric Singer's "VideoIn" and associated objects for MAX, "Big Eye" and David Rokeby's "Very Nervous System" (VNS) are three such systems. All these systems measure the variation in light that is produced when objects are moving in space.

There is also a wide variety of sensing devices and systems that enable the capture of data from purely gestural movement as opposed to topographical positioning. MidiDancer, created by Mark Coniglio, captures the flexion of joints (wrists, elbows, hips and knees). The Miburi bodysuit, developed by Yamha similarly captures flexion in a number of joints and has pressure sensitive pads that attach to the soles of the feet and a keyboard-like device that fits on each hand that can send both on/off information and thumb controls that can send a range of voltages to be interpreted as pitch-bend or other MIDI controller type information. Infusion Systems makes ICube, an analogue to digital converter that accepts a variety of transducers and actuators like light, touch, flexion, thermal and proximity sensors, switches, L.E.D's and motors. SensorLab, created at STEIM is another analogue to MIDI interface for the prototyping of musical instruments and interactive control systems. And BodySynth, developed by Chris van Raalte uses electromyographic sensors to generate MIDI information from the contraction and relaxation of muscles.

There is also another category of gestural devices that were initially intended as MIDI controller instruments for musicians rather than for their potential as controllers for dancers, but may still have useful application in this area. Pat Downs' Airdrums, the GoldBrick Interface for the Nintendo PowerGlove, and well before MIDI there were Theramins, to name a few.

- **Some examples of interactive works**

2.1 "Dark Around the Edges" and "Songs for the Body Electric"

Two works by Todd Winkler,Walter Ferrero and Gerry Girouard, *Dark Around the Edges* and *Songs for the Body Electric* use Rokeby's Very Nervous System as the medium for interaction between dancers, lighting, video and music. In these works the video input and therefore the performance space is divided into a number of regions.

"Each region has the ability to trigger a particular MIDI note or a series of notes, report continuous changes in movement, reconfigure the software, and start or stop a musical process." (Winkler, 1998).

Winkler is able to reconfigure response behaviours in the system by using a number of presets embedded in the software Max. So a single movement gesture in the same position within the performance space can produce a variety of outcomes at different times, though in the works mentioned they are used for different styles of dancing. On the one hand for movements of a precise robotic nature where the musical response is machine-like percussive sounds and on the other characterised by a more free-flowing style of dance that produces continuously evolving sound.

"In several sections of the works, physical gestures have an obvious and immediate impact on the quality of timbre by mapping VNS values to control DSP functions, such as filters, low frequency oscillators, and distortion algorithms. Further processing under MIDI control is available from the Ensoniq DP/4 processor. Several techniques may be combined, such as in "Gazelles" from *Songs for the Body Electric*, where location triggers play chords at a maximum rate of two per second, while speed effects a low pass filter and pitch bend." (Winkler, 1998).

In these works as well as others by Winkler, highly complex and variable mapping procedures have been successfully deployed to create remarkable works that integrate sound, movement and visuals in a way that only a few years ago were impossible. The success of the works is not just due to the emerging technology, but very much because of the lengthy collaborative process that lies behind them. A lot of energy was spent in research and experimentation with different set-ups to determine the optimum for both composer and choreographer.

2.2 Your Sky is Filled with Billboards of the Sky

In Lindsay Vickery's *Your Sky is Filled with Billboards of the Sky* for dancer, video and computer generated sound, the mediating device for movement is the body suit hooked up to a Macintosh PowerBook and a Mac' G4. The PowerBook runs STEIM's *Image/ine* software while the G4 runs IRCAM/Cycling 74's MAX/MSP.

MAX/MSP takes note number, velocity, pitchbend, modulation and elapsed time between events from the MIBURI. Note numbers are mapped to samples, velocity to sample playback speed (pitch), controller 1 to panning, pitchbend to sample volume and if the time between events is greater than zero a loop of the previous few seconds is turned on[1].

Similar procedures are followed to control the video images that include a QuickTime video of clouds passing over the moon, a live video feed of the dancer, text and a still image of a graffitied wall.

For example, note numbers control the angle of displacement of the still image, velocity determines the 'x' coordinate of the still, pitchbend determines the 'y' coordinate, controller 1 alters the area of the image that is displayed, its hue, saturation, colour-shift and the level of "keyer" effect[2].

Various sensors allow the dancer to distort her image with 'PhotoShop'-like effects.

Your Sky is Filled with Billboards of the Sky was developed for the REV festival at the Brisbane Powerhouse in April 02 with Katherine Duhiggt performing. I was fortunate to see two performances during the Totally Huge New Music Festival in Perth this year with visiting artist Zhang Ping in the suit. Vickery states that Zhang Ping's dance, which is strongly influenced by Chinese classical, folk and modern dance, was particularly well suited to the piece. She structured her movements around expansion and contraction of the kinesphere (Laban & Ullman, 1966) with particular emphasis on the arms and torso as the prime articulators of transition from and expression within one extreme to the other.

[1] From conversations and information supplied by the composer.

[2] The amount of transparency applied to whiteness that allows a background image to show through.

A kind of slow, abstracted bird-like movement of the arms was a strong motif in these performances and was very effective in managing the panning of sounds and the coordinates of the image buffer.

So it was a fortunate collaboration that was not made easier by the language barrier. But even so Vickery laments the limited number of rehearsals that were available for the dancers to familiarise themselves with the controls.

2.3 MAP1

MAP1 by Garth Payne is an audience interactive installation that again uses VNS to collect movement data in the environment.

"This installation focuses on sound using the immersive, fluid and emotive qualities of the medium to generate a rich, enveloping and ever evolving environment. The sounds are made more fluid by the use of a system capable of moving the apparent source of the sound through the physical environment. This creates a dynamic relationship between the presence and position of a body and the position and movement trajectory of the sounds. The ability to move sound through space affords the sound a physicality and in so doing the sense that the sound becomes another physical character or presence within the installation.

Those within the installation will sense a physical interaction with the sound. A wide range of different aural qualities are mapped in qualitative groupings to different regions within the installation space, generating a plethora of aural textures and densities, chosen on the basis of the quality of movement of the body within that region of the exhibition" (Payne, 1998)

In the above examples, a number of paradigms are apparent in terms of performer control of the various media. In the first example, it is the extensive collaborative process that allows for the level of complexity exhibited. The choreographers at least if not all of the dancers spent a considerable time familiarising themselves with the technology and potential sonic responses.

Your Sky is Filled with Billboards of the Sky I think takes a pragmatic approach so that a successful outcome is achieved in a relatively short rehearsal period due to the compositional processes inherent in the piece and by limiting the parameters of control over the visual. Having said that though, the performer in this case still had a fearsome array of controls to think about while making a satisfactory choreography.

In the last example a more pedagogical approach is taken that necessarily involves a certain amount of trial and error. Participant interactions become more fluent over time with intent gradually matching outcome.

In many senses we reinvent the wheel each time we make a new movement-interactive piece, drawing up new maps that make various correspondences between (human) movement and sound. We look at movement and make decisions about how it might translate into sound. We make a sound and try to figure out a way to move in a sensitised space that might recreate that sound object. We look at a space and start chopping it up into little bits that can be managed by the technology. We ask performers or participants to take responsibility for completing the work in an area in which they have little expertise. In saying this I place no particular value or judgement on these decisions or methods, but it does raise the question of whether there might be some generalisations that we can make about correlating sound, space and movement and whether certain solutions might be more transportable from one situation to another. And not just physical situations, but social, technological, aesthetic and temporal.

One of the main objections to simple one-to-one relationships where for example a certain gesture will always result in the same sonic response is that it quickly becomes boring and predictable.

Virgil Thompson, composer of many works for dance, age 90 in an interview with Katherine Teck. "Music which illustrates the dance is just as boring as dance which illustrates the music." . . .

"Music walks down the street beside the dance". (Teck, 1989.p 43)

Norma Reynolds Dalby - "Furthermore she pointed out how very exciting it can be to find out 'what can happen in terms of enhancing both media if you do not follow the dancer precisely'". (Teck, 1989.p 53)

And Donald York - "But if the music and dance breathe independently and not exactly hand in glove, it is exciting." (Teck, 1989.p 71)

The music should never 'Mickey Mouse' the movement so we are told and yet we strive for an ever-greater intimacy between the two in the systems that we build. How to reconcile these seemingly conflicting ideas. My current research includes a

search for a generalised map that is readily adaptable to a range of spacial situations and dance styles.

Firstly a paradigm that might be useful is one in which the dancer is free to go about making a structure as freely as possible. Free from the responsibility of having to look after a massive number of musical interrelations and controls. Free from the necessity for extensive collaboration. A paradigm in which the sound/music has both elements of unison and elements of parallelism to the dance.

Some hunches I am following include making broad generalisations about space, sound and movement. Then building a system where these generalisations can be linked to see which might yield the best results across the widest range of possibilities.

3 One Map?

Before we are able to correlat e movement to sound we must have some generalised concept of space and a generalised relationship to sound. Ignoring the social, cultural or political sphere that any space might inhabit for a moment, let us assume that it is a completely empty space. A space devoid of objects or meaning except that it has boundaries like a floor, ceiling and walls or just open ground. A space in which any object placed in it will affect how we see that space and through our sensing system affect what we hear.

So what is the most obvious correlation between an empty space and sound? Silence surely. And if I were to fill that space entirely the result in sound would be white noise. If I was particularly perverse I could reverse the situation and say that the correlate of empty space is white noise and the full space silence. Either way a partial fullness of space, i.e. an object in the space is the equivalent of a band stop or band pass filter.

Of course we could substitute some other sub-set of white noise to represent fullness or the material with which a space is filled. All the audible tones of a chromatic scale for example or some other aggregate of tones or all the samples in our sound sample library.

If one were to build a single map that might serve for more than one dance, movement or installation work what parameters of sound would be mapped to the spacial dimensions of height, depth and width? If I used frequency, amplitude and disposition as the three primary elements of sound that a person moving in space controls, which axis applies to each of these elements?

The possibilities are:

1. X=frequency, Y=amplitude, intersection of XYZ=disposition
2. X=frequency, Z=amplitude, intersection of XYZ=disposition
3. Y=frequency, Z=amplitude, intersection of XYZ=disposition
4. X=amplitude, Y=frequency, intersection of XYZ=disposition
5. Y=amplitude, Z=frequency, intersection of XYZ=disposition
6. X=amplitude, Z=frequency, intersection of XYZ=disposition.

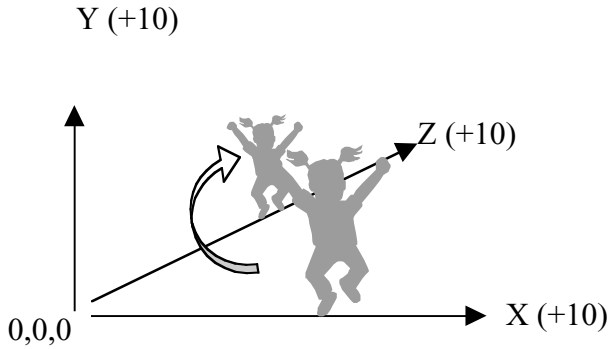

Figure 1. X = width, Y = height, Z = depth.

So what are we likely to hear in each of these possibilities?

1. In 1: As an object moves from co-ordinates X5, Y5, Z0 to X0, Y5, Z10 or in staging terms: down stage centre to up stage left (maintaining the same height all the way) we hear a fall in pitch and a steady volume while the sound travels from centre rear to left front. (Assuming here that there is a four-speaker placement with the observer in the centre of the four speakers.)
2. Using the same movement as above 2. gives us an increase in volume and the same fall in pitch as in 1. The same spacial distribution of sound applies to all the possibilities.
3. gives us a steady pitch and an increase in volume.
4. a decrease in volume from medium to very soft over a steady pitch.
5. a steady volume and a rise in pitch from one extreme to the other.

6. a decrease in volume from medium to very soft over a rise in pitch from one extreme to the other.

Now when we are dealing with an object in relation to a camera position the above results are modified. If the camera is positioned at, for example, a performer's eye level and in the first row of the audience, the performer fills more of the lens the closer she gets. So even when the height of the performance is static, but there is movement in the Z or depth axis the Y-axis will read an increase or decrease. There may also be some variation in the X axis as a receding object appears to our camera lens to get slightly thinner. (We'll ignore this as a marginal variation for the time being. Dancers generally don't need to look any thinner!) The above movement from down stage centre to up stage left gives us:

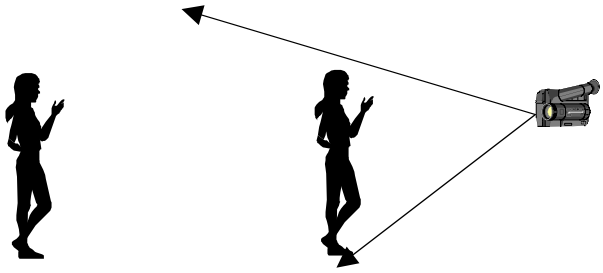

Figure 2. Objects closer to the lens fills more of the screen.

1. A decrease in volume and a fall in pitch while the sound travels from **high-to-low** centre rear to left front **medium-to-low**.
2. Is the same except for the alteration in distribution that applies to all possibilities.
3. gives us a fall in higher pitches and a rise in lower frequencies and an increase in volume.
4. a decrease in volume from medium to very soft over a fall in higher pitches and a rise in lower frequencies.
5. a fall in volume and a rise in pitch from one extreme to the other.
6. a decrease in volume from medium to very soft over a rise in pitch from one extreme to the other.

The above must be further qualified because we are dealing with a 3D object. So when I am talking about a rise or fall in pitch I am not talking about a single pitch but rather a band of frequencies and it is the whole band that rises or falls. And in relation to frequencies mapped to the Y axis (height) the band will be narrower or broader depending on where the object is on the Z axis.

So far I have been talking about just 3 sound parameters and the choice of frequency, amplitude and disposition might seem arbitrary. I think there are advantages in this choice. In particular the advantage of clarity and transparency, but other choices are available if we want to be less obvious. I could substitute resonance, or some other signal processing, for disposition resulting in:

H	X	A	A	F	F	R	R
H	Y	F	R	R	A	A	F
H	Z	R	F	A	R	F	A

Figure 3. Where A= amplitude, F = frequency and R = resonance

And so far I have not said much about the temporal domain. Speed of movement or tempo is another obvious element that should be built into the above permutations. I am currently playing with tempo as a means of introducing elements that go beyond the immediate one to one connections to movement. At higher rates of change in movement a slowed-down sonic "after-image" is triggered. Basically this is an isorhythmic canon and the idea behind it is essentially that the ear has another chance to catch the sound at a slower pace while also introducing some polyphony where there is just a solo dancer.

4 Conclusion

At this stage, without having completed and applied a fuller set of maps, it is not possible to say whether some of these generalisations will bear fruit for wider application even in just my own work. It is still a work in progress, but it is my belief though that more research into "obvious" correlations needs to be carried out so that more manifestations of artists' imagination have greater chance of making meaningful connections between collaborators, participants and audiences.

With the few example maps that have been experimented with, the results are encouraging. Even continuous unison of sound with movement has not proven to be so boring. (Certainly for the minimally inclined it can already be said to have been fruitful).

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